

Experiment 'p_6_7_reverse_array_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_7_reverse_array_seq

Natural language statement: null

Method signature: p_6_7_reverse_array_seq(inputs: seq<int>) returns (reversed: seq<int>)

Ensures

- `|reversed| == |inputs|`
- `forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_7_reverse_array_seq(inputs: seq<int>) returns (reversed: seq<int>)`

The method should respect the following contract:

`ensures |reversed| == |inputs|, ensures forall i :: 0 <= i < |reversed| ==> reversed[i]`
 ↪ `== inputs[|inputs| - 1 - i]`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnOOPYrZkzYYRhv9DacR0OLcfilvQ

Model: gpt-5.5-2026-04-23

Created at: 1780664209

Finish reason: stop

Usage: 318 tokens in, and 1387 tokens out

```
{
  reversed := [];
  var i := |inputs|;
  while i > 0
    invariant 0 <= i <= |inputs|
    invariant |reversed| == |inputs| - i
    invariant forall j :: 0 <= j < |reversed| ==> reversed[j] == inputs[|inputs| - 1 - j]
    decreases i
  {
    var prev := reversed;
    var oldI := i;
    i := i - 1;
    reversed := prev + [inputs[i]];

    assert oldI == i + 1;
    assert |prev| == |inputs| - oldI;
    assert |prev| == |inputs| - (i + 1);
    assert |prev| == |inputs| - 1 - i;
```

```

assert forall j :: 0 <= j < |reversed| ==> reversed[j] == inputs[|inputs| - 1 - j] by
  ↪ {
    forall j | 0 <= j < |reversed|
      ensures reversed[j] == inputs[|inputs| - 1 - j]
    {
      if j < |prev| {
        assert reversed[j] == prev[j];
        assert prev[j] == inputs[|inputs| - 1 - j];
      } else {
        assert j == |prev|;
        assert |inputs| - 1 - j == i;
        assert reversed[j] == inputs[i];
      }
    }
  }
}

```

Final Program

```

method p_6_7_reverse_array_seq(inputs: seq<int>) returns (reversed: seq<int>)
  ensures |reversed| == |inputs|
  ensures forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]
  ↪ ]
{
  reversed := [];
  var i := |inputs|;
  while i > 0
    invariant 0 <= i <= |inputs|
    invariant |reversed| == |inputs| - i
    invariant forall j :: 0 <= j < |reversed| ==> reversed[j] == inputs[|inputs| - 1 - j]
    decreases i
  {
    var prev := reversed;
    var oldI := i;
    i := i - 1;
    reversed := prev + [inputs[i]];

    assert oldI == i + 1;
    assert |prev| == |inputs| - oldI;
    assert |prev| == |inputs| - (i + 1);
    assert |prev| == |inputs| - 1 - i;

    assert forall j :: 0 <= j < |reversed| ==> reversed[j] == inputs[|inputs| - 1 - j] by
      ↪ {
        forall j | 0 <= j < |reversed|
          ensures reversed[j] == inputs[|inputs| - 1 - j]
        {
          if j < |prev| {
            assert reversed[j] == prev[j];

```

```
        assert prev[j] == inputs[|inputs| - 1 - j];
    } else {
        assert j == |prev|;
        assert |inputs| - 1 - j == i;
        assert reversed[j] == inputs[i];
    }
}
}
}
}
```

Total Token Usage

Input tokens: 318

Output tokens: 1387

Reasoning tokens: 1024

Sum of ‘total tokens’: 1705

Experiment Timings

Overall Experiment started at 1780664208783, ended at 1780664260291, lasting 51508ms (51.51 seconds)

Iteration #1 started at 1780664208785, ended at 1780664259969, lasting 51184ms (51.18 seconds)